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Master of Computer Applications
Semester – III
Data Analytics for Business
Experiment No: 5
Title: Building Sales Performance Dashboards

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Building Sales Performance Dashboards

Github Link: <https://github.com/San-bytes/Data-Analytics-For-Business/tree/main/Walmart-Dashboard>

Problem Statement:

You are a Sales Analyst for a retail chain. The sales director wants to monitor revenue, profit, product performance, and regional sales through a single dashboard. Design an interactive sales dashboard that supports data-driven decision-making.

Methodology:

Data Collection

- The Walmart Sales dataset was downloaded from Kaggle.
- The dataset was imported into Microsoft Power BI using the Get Data option.
- The dataset was checked and prepared using Power Query.

Data Preparation

- Checked column names and data types.
- Converted Date column into Date format.
- Verified numerical columns such as Weekly Sales, Fuel Price, Temperature, CPI, and Unemployment.
- Checked for missing values and data errors.
- Applied transformations before creating visualizations.

Dashboard Development

An interactive Sales Performance Dashboard was created using Power BI.

The dashboard includes:

- Total Sales KPI Card
- Average Weekly Sales KPI Card
- Total Store Count KPI Card
- Holiday Week Analysis
- Store-wise Sales Analysis
- Sales Trend Over Time
- Holiday Vs Non-Holiday Sales Comparison
- Fuel Price and Sales Relationship
- Temperature and Sales Relationship
- Interactive Slicers for Store, Date, and Holiday Flag

Dashboard Description

The dashboard provides an overall view of Walmart sales performance.

It helps users:

- Monitor total revenue generated from weekly sales.
- Compare sales performance among different stores.
- Analyze sales trends over time.
- Understand the effect of holidays on sales.
- Explore relationships between external factors and sales performance using interactive charts.

Dashboard Images

Image 1: Overall Sales Dashboard

- The dashboard displays Walmart's overall sales performance for the complete dataset from 2010 to 2012 without applying any filters.
- It shows total sales of 6.74bn, average weekly sales of 1.05M, 45 total stores, and 450 holiday weeks along with store-wise sales analysis, weekly sales trends, holiday versus non-holiday sales distribution, and a summary table.
- It provides a complete overview of business performance and supports effective sales analysis through interactive visualizations.

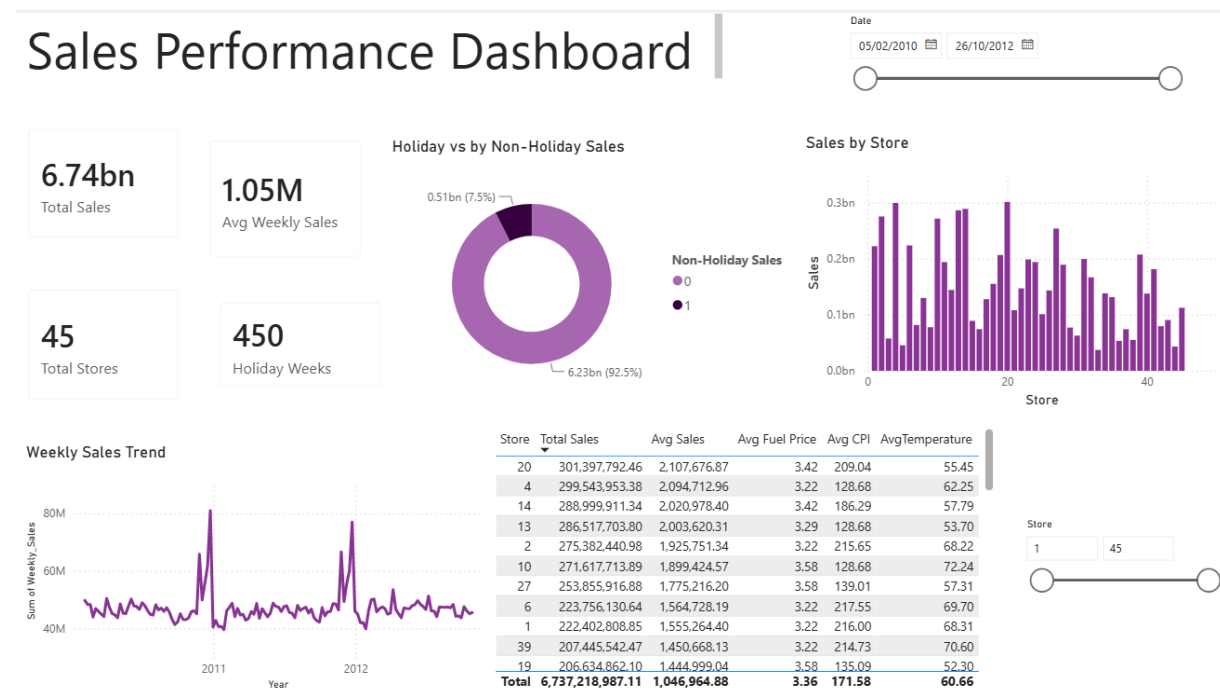


Image 2: Dashboard with Store Filter Selection

- After selecting stores 7 to 22 using the Store slicer, all dashboard visuals are automatically updated to display data for the selected stores only.
- The KPI cards change to total sales of 2.68bn, average weekly sales of 1.17M, 16 total stores, and 160 holiday weeks, while the charts and summary table display information related to the selected stores.
- This interaction helps compare the performance of selected stores and identify high-performing store groups.

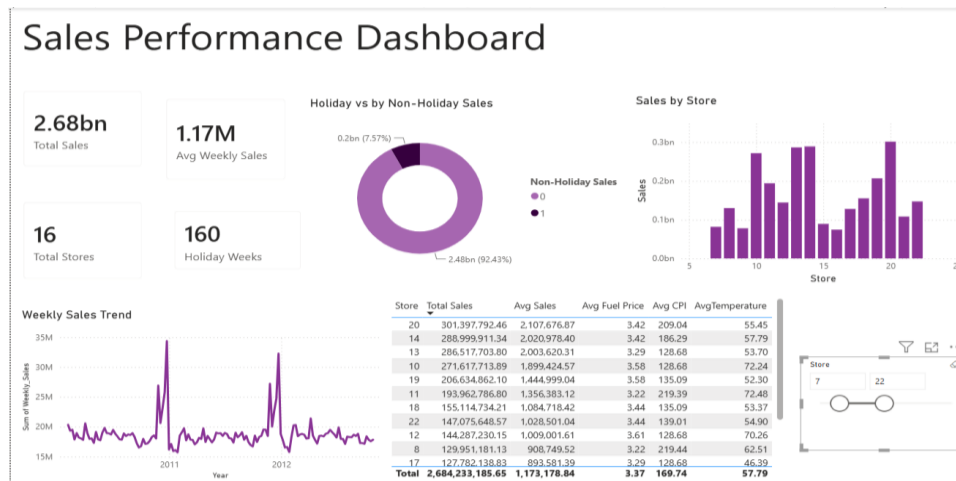
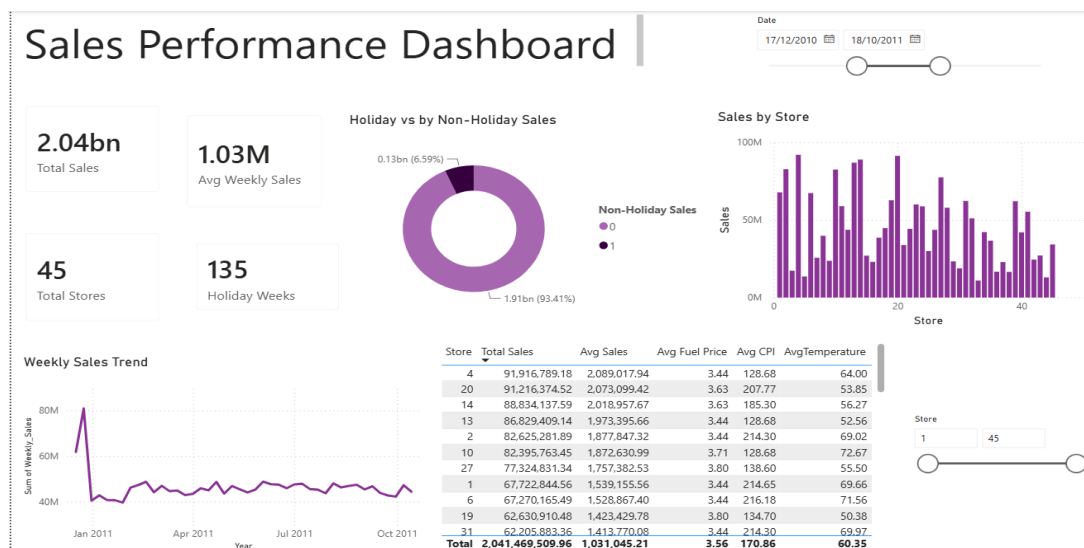


Image 3: Dashboard with Date Filter Selection

- After applying the date filter from 17-12-2010 to 18-10-2011, all dashboard visuals update to display data for the selected time period.
- The KPI cards show total sales of 2.04bn, average weekly sales of 1.03M, 45 total stores, and 135 holiday weeks, while the charts and summary table automatically reflect the filtered sales information.
- This interaction helps compare sales performance across different time periods and identify weekly sales trends more effectively.



Business Insights

- Walmart generated total weekly sales of approximately **6.74 billion** across all stores.
- The dataset contains performance records for **45 stores**.
- Sales values vary significantly between different stores, helping identify better-performing locations.
- Holiday periods contribute to changes in sales patterns.
- Sales trends show fluctuations over time with noticeable peaks during certain periods.
- External factors such as fuel price, temperature, CPI, and unemployment can be compared with sales performance.

Conclusion

The Walmart Sales dataset was successfully imported and analyzed using Microsoft Power BI. An interactive Sales Performance Dashboard was created using KPI cards, charts, and slicers, the dashboard helps analyze store performance, sales trends, and business factors affecting sales and the interactive features allow users to filter and explore data from different perspectives, supporting effective business decision-making.